

GIA 09.02.06 — SAFE MANUAL (DEBIAN 13 + PNET)

Версия БЕЗ опасного `cat > file`.

Принцип:

- открываем конфиг через nano
- ДОБАВЛЯЕМ строки
- НЕ удаляем дефолтный конфиг Debian
- НЕ ломаем include/source/systemd

===== 0. БАЗОВАЯ ПРОВЕРКА =====

```
# COPY/PASTE
```

```
ip -br a  
hostname
```

===== 1. ISP =====

```
# COPY/PASTE
```

```
hostnamectl set-hostname isp.au-team.irpo  
nano /etc/network/interfaces
```

Добавить НИЖЕ existing строк:

```
# MANUAL INPUT
```

```
auto ens3  
iface ens3 inet dhcp  
  
auto ens4  
iface ens4 inet static  
address 172.16.1.1/28  
  
auto ens5  
iface ens5 inet static  
address 172.16.2.1/28
```

НЕ УДАЛЯТЬ:

```
# MANUAL INPUT
source /etc/network/interfaces.d/*
auto lo
iface lo inet loopback
```

```
# COPY/PASTE
systemctl restart networking
```

```
# COPY/PASTE
nano /etc/sysctl.d/99-router.conf
```

Вставить:

```
# MANUAL INPUT
net.ipv4.ip_forward=1
```

```
# COPY/PASTE
sysctl --system
```

```
# COPY/PASTE
nano /etc/nftables.conf
```

НЕ УДАЛЯТЬ existing:

```
# MANUAL INPUT
flush ruleset
table inet filter
```

Добавить В КОНЕЦ:

```
# MANUAL INPUT
table ip nat {
  chain postrouting {
    type nat hook postrouting priority 100;
    oifname "ens3" masquerade
  }
}
```

```
# COPY/PASTE
systemctl restart nftables
ping -c 2 8.8.8.8
```

===== 2. HQ-RTR =====

```
# COPY/PASTE
hostnamectl set-hostname hq-rtr.au-team.irpo
nano /etc/network/interfaces
```

Добавить:

```
# MANUAL INPUT
auto ens3
iface ens3 inet static
    address 172.16.1.2/28
    gateway 172.16.1.1

auto ens4
iface ens4 inet static
    address 192.168.100.1/27

auto ens5
iface ens5 inet static
    address 192.168.100.33/28
```

```
# COPY/PASTE
systemctl restart networking
```

```
# COPY/PASTE
nano /etc/sysctl.d/99-router.conf
```

Вставить:

```
# MANUAL INPUT
net.ipv4.ip_forward=1
```

```
# COPY/PASTE
sysctl --system
```

```
# COPY/PASTE
nano /etc/nftables.conf
```

Добавить В КОНЕЦ:

```
# MANUAL INPUT
table ip nat {
  chain prerouting {
    type nat hook prerouting priority 0;
    iifname "ens3" tcp dport 8080 dnat to 192.168.100.2:8080
    iifname "ens3" tcp dport 2026 dnat to 192.168.100.2:2026
  }

  chain postrouting {
    type nat hook postrouting priority 100;
    oifname "ens3" masquerade
  }
}
```

```
# COPY/PASTE
systemctl restart nftables
ping -c 2 8.8.8.8
```

===== 3. HQ-SRV =====

```
# COPY/PASTE
hostnamectl set-hostname hq-srv.au-team.irpo
nano /etc/network/interfaces
```

Добавить:

```
# MANUAL INPUT
auto ens3
iface ens3 inet static
  address 192.168.100.2/27
  gateway 192.168.100.1
  dns-nameservers 8.8.8.8
```

```
# COPY/PASTE
systemctl restart networking
ping -c 2 8.8.8.8
```

===== 4. HQ-CLI =====

```
# COPY/PASTE
hostnamectl set-hostname hq-cli.au-team.irpo
nano /etc/network/interfaces
```

Добавить:

```
# MANUAL INPUT
auto ens3
iface ens3 inet static
    address 192.168.100.34/28
    gateway 192.168.100.33
```

```
# COPY/PASTE
systemctl restart networking
ping -c 2 8.8.8.8
```

===== 5. BR-RTR =====

```
# COPY/PASTE
hostnamectl set-hostname br-rtr.au-team.irpo
nano /etc/network/interfaces
```

Добавить:

```
# MANUAL INPUT
auto ens3
iface ens3 inet static
    address 172.16.2.2/28
    gateway 172.16.2.1

auto ens4
iface ens4 inet static
    address 192.168.200.1/28
```

```
# COPY/PASTE
systemctl restart networking
```

```
# COPY/PASTE
nano /etc/sysctl.d/99-router.conf
```

Вставить:

```
# MANUAL INPUT
net.ipv4.ip_forward=1
```

```
# COPY/PASTE
sysctl --system
```

```
# COPY/PASTE
nano /etc/nftables.conf
```

Добавить:

```
# MANUAL INPUT
table ip nat {
  chain prerouting {
    type nat hook prerouting priority 0;
    iifname "ens3" tcp dport 8080 dnat to 192.168.200.2:8080
    iifname "ens3" tcp dport 2026 dnat to 192.168.200.2:2026
  }

  chain postrouting {
    type nat hook postrouting priority 100;
    oifname "ens3" masquerade
  }
}
```

```
# COPY/PASTE
systemctl restart nftables
ping -c 2 8.8.8.8
```

===== 6. BR-SRV =====

```
# COPY/PASTE
hostnamectl set-hostname br-srv.au-team.irpo
nano /etc/network/interfaces
```

Добавить:

```
# MANUAL INPUT
auto ens3
iface ens3 inet static
    address 192.168.200.2/28
    gateway 192.168.200.1
    dns-nameservers 8.8.8.8
```

```
# COPY/PASTE
systemctl restart networking
ping -c 2 8.8.8.8
```

===== 7. УСТАНОВКА ПАКЕТОВ =====

```
# COPY/PASTE
apt update
```

```
# COPY/PASTE
apt install -y sudo curl wget vim nano net-tools tcpdump nftables dnsutils
traceroute iproute2 openssh-server chrony vlan frr isc-dhcp-server dnsmasq
nginx apache2-utils docker.io docker-compose apache2 mariadb-server php php-
mysql samba winbind libnss-winbind krb5-user smbclient realmd sssd sssd-
tools libnss-sss libpam-sss adcli packagekit
```

===== 8. SSH =====

HQ-SRV + BR-SRV

```
# COPY/PASTE
nano /etc/ssh/sshd_config
```

Добавить В КОНЕЦ:

```
# MANUAL INPUT
Port 2026
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 2
```

```
Banner /etc/ssh_banner  
AllowUsers sshuser
```

НЕ УДАЛЯТЬ:

```
# MANUAL INPUT  
Subsystem sftp  
Include /etc/ssh/ssh_config.d/*.conf
```

```
# COPY/PASTE  
echo 'Authorized access only' > /etc/ssh_banner  
useradd -m -s /bin/bash sshuser  
echo 'sshuser:P@ssw0rd' | chpasswd  
systemctl restart ssh
```

===== 9. GRE =====

HQ-RTR

```
# COPY/PASTE  
ip tunnel add gre1 mode gre local 172.16.1.2 remote 172.16.2.2 ttl 64  
ip addr add 10.10.0.1/30 dev gre1  
ip link set gre1 up
```

BR-RTR

```
# COPY/PASTE  
ip tunnel add gre1 mode gre local 172.16.2.2 remote 172.16.1.2 ttl 64  
ip addr add 10.10.0.2/30 dev gre1  
ip link set gre1 up
```

===== 10. FRR + OSPF =====

HQ-RTR + BR-RTR

```
# COPY/PASTE  
nano /etc/frr/daemons
```

Изменить:


```
# MANUAL INPUT
ospfd=yes
```

```
# COPY/PASTE
systemctl restart frr
```

HQ-RTR

```
# COPY/PASTE
vtysh
```

Вставить:

```
# MANUAL INPUT
conf t
router ospf
  router-id 1.1.1.1
  network 192.168.100.0/27 area 0
  network 192.168.100.32/28 area 0
  network 10.10.0.0/30 area 0
exit
interface gre1
  ip ospf authentication
  ip ospf authentication-key password
  ip ospf network point-to-point
exit
write
```

BR-RTR

```
# COPY/PASTE
vtysh
```

Вставить:

```
# MANUAL INPUT
conf t
router ospf
  router-id 2.2.2.2
  network 192.168.200.0/28 area 0
  network 10.10.0.0/30 area 0
exit
interface gre1
  ip ospf authentication
```

```
ip ospf authentication-key password
ip ospf network point-to-point
exit
write
```

===== 11. DHCP =====

HQ-RTR

```
# COPY/PASTE
nano /etc/default/isc-dhcp-server
```

Изменить:

```
# MANUAL INPUT
INTERFACESv4="ens5"
```

```
# COPY/PASTE
nano /etc/dhcp/dhcpd.conf
```

Добавить В КОНЕЦ:

```
# MANUAL INPUT
default-lease-time 600;
max-lease-time 7200;

option domain-name "au-team.irpo";
option domain-name-servers 192.168.100.2;

subnet 192.168.100.32 netmask 255.255.255.240 {
    range 192.168.100.34 192.168.100.46;
    option routers 192.168.100.33;
}
```

```
# COPY/PASTE
systemctl restart isc-dhcp-server
```

===== 12. DNS =====

HQ-SRV

```
# COPY/PASTE
nano /etc/dnsmasq.conf
```

Добавить В КОНЕЦ:

```
# MANUAL INPUT
interface=ens3
listen-address=127.0.0.1,192.168.100.2
bind-interfaces
server=8.8.8.8
domain=au-team.irpo
address=/hq-rtr.au-team.irpo/192.168.100.1
address=/hq-srv.au-team.irpo/192.168.100.2
address=/hq-cli.au-team.irpo/192.168.100.34
address=/br-rtr.au-team.irpo/192.168.200.1
address=/br-srv.au-team.irpo/192.168.200.2
```

```
# COPY/PASTE
systemctl restart dnsmasq
```

===== 13. DOCKER =====

BR-SRV

```
# COPY/PASTE
docker load -i /root/Additional/docker/site_latest.tar
docker load -i /root/Additional/docker/mariadb_latest.tar
mkdir -p /root/testapp
nano /root/testapp/docker-compose.yml
```

Вставить:

```
# MANUAL INPUT
version: '3'
services:
  db:
    image: mariadb_latest
    environment:
```

```
MYSQL_ROOT_PASSWORD: P@ssw0rd
```

```
site:  
  image: site_latest  
ports:  
  - "8080:80"
```

```
# COPY/PASTE
```

```
cd /root/testapp  
docker compose up -d
```

===== 14. APACHE =====

HQ-SRV

```
# COPY/PASTE
```

```
mysql -e "CREATE DATABASE webdb;"  
mysql -e "CREATE USER 'web'@'localhost' IDENTIFIED BY 'P@ssw0rd';"  
mysql -e "GRANT ALL PRIVILEGES ON webdb.* TO 'web'@'localhost';"  
mysql -e "FLUSH PRIVILEGES;"
```

```
# COPY/PASTE
```

```
mysql webdb < /root/Additional/web/dump.sql  
cp /root/Additional/web/index.php /var/www/html/
```

```
# COPY/PASTE
```

```
nano /etc/apache2/ports.conf
```

Изменить:

```
# MANUAL INPUT
```

```
Listen 8080
```

```
# COPY/PASTE
```

```
nano /etc/apache2/sites-available/000-default.conf
```

Изменить:

```
# MANUAL INPUT
<VirtualHost *:8080>
```

```
# COPY/PASTE
systemctl restart apache2
```

===== 15. NGINX =====

ISP

```
# COPY/PASTE
htpasswd -bc /etc/nginx/.htpasswd WEB P@ssw0rd
nano /etc/nginx/sites-available/proxy
```

Вставить:

```
# MANUAL INPUT
server {
    listen 80;
    server_name web.au-team.irpo;

    auth_basic "Restricted";
    auth_basic_user_file /etc/nginx/.htpasswd;

    location / {
        proxy_pass http://172.16.1.2:8080;
    }
}

server {
    listen 80;
    server_name docker.au-team.irpo;

    location / {
        proxy_pass http://172.16.2.2:8080;
    }
}
```

```
# COPY/PASTE
ln -sf /etc/nginx/sites-available/proxy /etc/nginx/sites-enabled/proxy
rm -f /etc/nginx/sites-enabled/default
```

```
nginx -t
systemctl restart nginx
```

===== 16. SAMBA DC =====

BR-SRV

```
# COPY/PASTE
systemctl stop smbd nmbd winbind samba-ad-dc
mv /etc/samba/smb.conf /etc/samba/smb.conf.bak
```

```
# COPY/PASTE
samba-tool domain provision --realm=AU-TEAM.IRPO --domain=AU-TEAM --server-
role=dc --dns-backend=SAMBA_INTERNAL --adminpass='P@ssw0rd'
```

```
# COPY/PASTE
cp /var/lib/samba/private/krb5.conf /etc/krb5.conf
systemctl unmask samba-ad-dc
systemctl enable samba-ad-dc
systemctl restart samba-ad-dc
```

===== 17. DOMAIN USERS =====

```
# COPY/PASTE
samba-tool group add hq
```

```
# COPY/PASTE
for i in 1 2 3 4 5; do samba-tool user create hquser$i P@ssw0rd; done
```

```
# COPY/PASTE
samba-tool group addmembers hq hquser1,hquser2,hquser3,hquser4,hquser5
```

===== 18. JOIN DOMAIN =====

HQ-CLI

```
# COPY/PASTE
echo 'nameserver 192.168.200.2' > /etc/resolv.conf
realm join -U Administrator au-team.irpo
```

Пароль:

```
# MANUAL INPUT
P@ssw0rd
```

```
# COPY/PASTE
pam-auth-update --enable mkhomedir
realm deny --all
realm permit -g hq@au-team.irpo
```

===== 19. CHRONY =====

ISP

```
# COPY/PASTE
nano /etc/chrony/chrony.conf
```

Добавить:

```
# MANUAL INPUT
local stratum 5
allow 0.0.0.0/0
```

HQ

```
# COPY/PASTE
nano /etc/chrony/chrony.conf
```

Добавить:

```
# MANUAL INPUT
server 172.16.1.1 iburst
```

BR

```
# COPY/PASTE
nano /etc/chrony/chrony.conf
```

Добавить:

```
# MANUAL INPUT
server 172.16.2.1 iburst
```

```
# COPY/PASTE
systemctl restart chrony
```

===== 20. ПРОВЕРКИ =====

```
# COPY/PASTE
vtysh -c "show ip ospf neighbor"
```

```
# COPY/PASTE
ip r
```

```
# COPY/PASTE
dig br-srv.au-team.irpo
```

```
# COPY/PASTE
curl http://web.au-team.irpo
```

```
# COPY/PASTE
docker ps
```

```
# COPY/PASTE
realm list
```



```
# COPY/PASTE  
chronyc sources
```

```
# COPY/PASTE  
systemctl --failed
```